

Patient Shortage in Histopathology: Is the Residual Breadth Benefit a Site-Coverage Benefit?

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September 11, 2026

Abstract

At a fixed patch budget, TCGA-UT patch classifiers trained on more patients are more accurate, and adjusting for within-patient redundancy across all feature directions still leaves a benefit of 2.62 percentage points per doubling of patients. This experiment tests whether that residual reflects coverage of tissue source sites, the institutions that contributed TCGA tissue. Three nested training allocations share a budget of 160 patches per class: five patients from five sites with 32 patches each, twenty patients from the same five sites with eight patches each, and twenty patients from ten sites with eight patches each. The first step adds patients without adding sites; the second adds sites without changing the number of patients or patches. Thirteen of the 30 cancer types have enough patients per site to enter the contrast. For these types, doubling the number of sites raises patient-macro balanced accuracy by 1.06 percentage points (test-patient 95% interval: 0.74 to 1.37). Quadrupling patients within fixed sites leaves a benefit beyond effective support of 3.31 points per doubling of patients (2.71 to 3.84), but most of this value reflects a deep site allocation that is 3.70 points less accurate than randomly drawn five-patient allocations; on the breadth–depth grid, the residual for these types is 1.20 points. Formally, the within-site residual exceeds the one-point practical threshold, whereas the site gain is not clearly practically relevant and is positive in only eleven of fifteen training draws; under the prespecified criteria, the outcome is inconclusive. The site gain falls mainly on test patients from the added sites, at 3.93 points, and amounts to 0.67 points for patients from sites absent from both allocations. For the larger cancer types that permit this test, restricting twenty patients to five institutions costs no accuracy, and the site gain is more consistent with matching of seen sites than with robustness to unseen ones.

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1 Motivation

How a labelling budget is spread over patients matters more than its size. At 160 patches per class on TCGA-UT, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 9.52 percentage points of patient-macro balanced accuracy (Queisler, 2026a). Part of this advantage is redundancy: patches of one patient repeat each other, so they are worth fewer independent observations than their count suggests. *Effective support* discounts the patch count for this redundancy through an intraclass correlation, the share of feature variance that lies between patients. Even with a correlation computed across all 2,560 feature directions, each doubling of patients still adds 2.62 points at equal effective support (95% interval: 2.04 to 3.17), and the better redundancy measure removes only about 12% of the residual (Queisler, 2026b). Underestimated redundancy is therefore an unlikely explanation. Additional patients supply information that effective support does not count.

Two sources of such information are plausible. TCGA-UT slides were contributed by many *tissue source sites*, the hospitals and biobanks that collected and prepared the tissue. Sites differ in tissue handling, staining, and patient population, and deep models recognize these site-specific signatures in TCGA slides (Howard et al., 2021). A classifier trained on patients from few sites may learn cancer types under a narrow set of site conditions and transfer poorly to other sites. Alternatively, patients differ biologically even within one institution: additional patients may contribute morphological variants of a cancer type that repeated crops of a few uniform tumour regions cannot supply (Komura et al., 2022). Both accounts predict a benefit of breadth beyond effective support. They differ in one testable respect: under site coverage, additional patients help mainly when they come from sites not yet represented.

Existing allocations cannot separate the two accounts. When patients are drawn at random, adding patients also adds sites, so any quantity computed from the breadth–depth grid moves together with patient count. Separating them requires an intervention that changes the number of sites while the number of patients and patches stays fixed, and that changes the number of patients while the sites stay fixed. Sites are the natural first candidate for such an intervention because they are recorded for every TCGA participant, whereas the dataset provides no labels for morphological subtypes. The experiment is informative in both directions: if new sites add nothing, a breadth benefit that persists within fixed sites localizes the missing information in variation between patients of the same institution.

The question is therefore: *does a training patient from a new tissue source site contribute more than an additional patient from a site that is already represented, and do additional patients still help beyond effective support when they add no new sites?*

2 Study design

Only the composition of the training allocations changes. The design inherits from Queisler (2026a) the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features formed by concatenating the class token and the mean patch token (Zimmermann et al., 2024), the three fixed patient-disjoint train–validation–test splits, and the natural validation and test partitions. It also inherits the patch eligibility rule, the nested patch ordering, and the read-out: multinomial logistic regression with its regularization strength selected on the validation partition separately for every fit. TCGA-UT participants are called patients throughout.

A patient’s site is the two-character tissue source site code in its TCGA barcode, which has the form TCGA-XX-YYYY with site code XX. A site thus stands for everything shared by the tissue of one contributing institution, including preparation, staining, and the local patient population. The design does not separate these components.

Table 1 lists the three allocations. Each one fixes the number of patients G , patches per patient m , and sites S for every class that enters the site contrast. The first step repeats the equal-budget

Allocation	Patients G	Patches per patient m	Sites S	Patients per site
Deep	5	32	5	1
Broad, five sites	20	8	5	4
Broad, ten sites	20	8	10	2

Table 1: The three allocations share a budget of 160 patches per class and form two steps. From the deep to the broad five-site allocation, patient count quadruples within the same sites. From the broad five-site to the broad ten-site allocation, site count doubles while patients, patches per patient, and budget stay fixed.

comparison of Queisler (2026a) under one restriction: the fifteen additional patients come from the five sites that the deep allocation already covers. The second step moves the same logic one level up the hierarchy of patches, patients, and sites. Where the breadth–depth grid exchanged patches of a known patient for new patients, this step exchanges patients of a known site for patients of new sites: ten of the twenty patients are replaced by ten patients from five further sites. Four patients per site is the smallest number that lets twenty patients come from the deep allocation’s five sites, and doubling the site count matches the per-doubling scale on which the residual breadth benefit is expressed.

2.1 Allocation rules

Three rules turn the allocations into concrete training sets. Together they ensure that each step changes only its intended factor and that the sites of both broad allocations are comparable.

The first rule defines a common site pool. For split r and class c , let $\mathcal{E}_{rc}$ contain the training patients with at least 32 patches of the class, as in Queisler (2026a), and let $\mathcal{E}_{rcs} \subseteq \mathcal{E}_{rc}$ contain those from site s . The site pool is

$$\mathcal{S}_{rc} = \{s : |\mathcal{E}_{rcs}| \geq 4\}. \quad (1)$$

A class enters the site contrast, and is called a *multi-site class*, only if $|\mathcal{S}_{rc}| \geq 10$ in all three splits. Every site of both broad allocations is drawn from this pool, so sites that are only represented in the ten-site allocation must be as large as those shared with the five-site allocation. Without this requirement, the five-site allocation would draw from the largest contributing institutions and the ten-site allocation would add smaller ones, so the step would change site size as well as site count.

The second rule nests sites and patients across the three allocations. For each split, multi-site class, and draw $d \in \{1, \dots, 5\}$, ten sites are drawn without replacement from $\mathcal{S}_{rc}$, and five of them are chosen at random as *core sites*; the other five are *added sites*. For each of the ten sites, an ordered list of four patients is drawn without replacement from $\mathcal{E}_{rcs}$. The broad five-site allocation uses all four listed patients of every core site, the deep allocation uses the first listed patient of every core site, and the broad ten-site allocation uses the first two listed patients of every core and every added site. Writing $\mathcal{P}^D$, $\mathcal{P}^5$, and $\mathcal{P}^{10}$ for the patient sets of the three allocations,

$$\mathcal{P}^D \subset \mathcal{P}^5, \quad |\mathcal{P}^5 \cap \mathcal{P}^{10}| = 10, \quad \mathcal{P}^D \subset \mathcal{P}^{10}. \quad (2)$$

Patches follow the nested round-robin order over each patient’s slides, so a patient contributes its first m patches. The eight patches of a deep patient in either broad allocation are therefore the first eight of its 32 deep patches. Classes are sampled independently, so a site that contributes to several cancer types can be drawn for each of them.

The third rule keeps all other classes fixed. Classes that do not meet the site-pool requirement remain in training, so the classifier always solves the same 30-class task. For each split and draw,

they receive one random allocation of ten patients contributing sixteen patches each, drawn as in Queisler (2026a) and shared by all three allocations. Every class thus keeps 160 patches, and only the training material of multi-site classes differs between allocations. Before any model is fitted, the number of multi-site classes per split is recorded. The experiment proceeds only if at least ten classes qualify, so that the site contrast rests on a substantial part of the label space; otherwise the site levels are redesigned against this census before any classifier is trained. The design requires 45 fits: three allocations, three splits, and five draws.

3 Evaluation

The evaluation asks two questions, one for each step of Table 1. Both use the same accuracy measure and the same uncertainty procedure.

3.1 Site gain and within-site residual breadth

Accuracy is patient-macro balanced accuracy restricted to multi-site classes: patch recall is averaged within each test patient and true class, then equally across patients within each class, and finally across the multi-site classes. An allocation score A , in percent, averages the three splits and five draws equally, giving A_D , A_5 , and A_{10} for the deep, five-site, and ten-site allocations. Balanced accuracy over all 30 classes is reported alongside.

The *site gain* compares the two broad allocations,

$$\Delta_S = A_{10} - A_5, \quad (3)$$

and measures what five additional sites contribute when patients, patches per patient, and budget stay fixed. Under site coverage, Δ_S is positive; if sites are irrelevant beyond the patients they supply, it is close to zero.

The first step cannot be read from the raw difference $A_5 - A_D$ alone, because quadrupling patients also raises effective support. With $\bar{\rho}_c$ the full-feature intraclass correlation of class c averaged over the three splits (Queisler, 2026b), effective support of class c under G patients contributing m patches each is

$$N_{\text{eff}}(G, m) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{Gm}{1 + (m-1)\bar{\rho}_c}, \quad (4)$$

averaged over the set $\mathcal{C}$ of multi-site classes. The part of the first step that effective support explains is $\hat{\beta} \log(N_{\text{eff}}(20, 8)/N_{\text{eff}}(5, 32))$, where $\hat{\beta}$ is the effective-support slope of the regression $A = \alpha + \beta \log N_{\text{eff}} + \gamma \log G$. This regression is refitted to the stored predictions of the nine-cell breadth–depth grid (Queisler, 2026a), with accuracy restricted to the same multi-site classes; no classifier is retrained for this purpose. Because the first step quadruples patient count, the remainder is divided by two doublings, giving the *within-site residual breadth benefit*

$$b_W = \frac{A_5 - A_D - \hat{\beta} \log(N_{\text{eff}}(20, 8)/N_{\text{eff}}(5, 32))}{2} \quad (5)$$

in percentage points per doubling of patients. The same refitted regression also gives a reference residual $b_{\text{ref}} = \hat{\gamma} \log 2$ for the multi-site classes, estimated from allocations in which sites were not controlled. If sites carry the residual breadth benefit, b_W falls towards zero while b_{ref} stays positive. If sites do not matter, b_W stays close to b_{ref} .

A secondary analysis asks whether a site gain reflects matching or diversity. For each draw, a test patient of a multi-site class is assigned to one of three strata by its site: a core site, an added site, or a site not drawn for that class. The site gain is then computed separately within each stratum, averaging patient recalls with their weights in the primary endpoint. *Site*

matching predicts a gain concentrated on patients from added sites, whose conditions the ten-site allocation has seen and the five-site allocation has not, together with a small loss on core-site patients, whose sites lose two of four training patients. *Site diversity* predicts a gain also for patients from sites drawn for neither allocation, the stratum closest to deployment at an unseen institution. Stratum sizes vary between draws, so these estimates are descriptive and carry no decision threshold.

3.2 Uncertainty

Test-patient uncertainty follows Queisler (2026a). Each of 2,000 paired Bayesian bootstrap replicates assigns $\text{Dirichlet}(1, \dots, 1)$ weights to the distinct test patients (Rubin, 1981). A patient’s weight is shared across allocations, splits, draws, classes, and repeated test appearances. Because the test partitions coincide with those of the breadth–depth grid, the same weights also enter the refitted regression, so $\hat{\beta}$, b_{ref} , and b_W are recomputed on every replicate and remain paired. Each stored distribution contains 2,001 entries: the observed estimate, computed with equal patient weights, followed by the 2,000 resampled estimates. The 2.5th and 97.5th percentiles of the resampled estimates, excluding the observed entry, give exploratory 95% intervals, including one for the difference $b_W - b_{\text{ref}}$. Correlations are held fixed at their point estimates. The dispersion of each allocation’s accuracy, of the site gain, and of the raw first step $A_5 - A_D$ across the fifteen split–draw combinations is reported as a descriptive measure of sensitivity to which sites and patients are drawn; it includes variation between splits and is not combined with the test-patient intervals.

4 Interpretation criteria

Conclusions rest on the test-patient intervals for Δ_S and b_W and on the practical threshold of one percentage point, used for both quantities as in Queisler (2026b). Table 2 summarizes the resulting readings.

Account	Claim	Site gain Δ_S	Within-site residual b_W
Site coverage	Additional patients help because they add institutions	Interval above 1	Interval within $[-1, 1]$
Within-site patient variation	Additional patients help even from the same institutions	Interval within $[-1, 1]$	Interval above 1
Both	Sites and patients each contribute	Interval above 1	Interval above 1

Table 2: Each account predicts a distinct combination of the two contrasts. Intervals are test-patient 95% intervals in percentage points; b_W is expressed per doubling of patients. All other combinations, including any interval that crosses a threshold, are inconclusive.

Site coverage. If new sites add a practically relevant gain and no residual remains within fixed sites, the missing source of information on TCGA-UT is coverage of institutions. The deficit would then be a second level of redundancy: patients of one site resemble each other, just as patches of one patient do. This reading would motivate an effective support that discounts

patches within patients and patients within sites, together with site-aware sampling or augmentation as a mitigation method. The stratified analysis indicates whether such a method should target matching of known sites or robustness to unseen ones.

Within-site patient variation. If new sites add nothing while the within-site residual stays above one point, the benefit of additional patients does not depend on institutional variety. The missing information would then lie in differences between patients of the same institution, most plausibly biological variation within a cancer type. A signal for this shortage would need to measure how much of a class’s morphological range the training patients cover, rather than which sites they come from.

Both and inconclusive outcomes. If both contrasts exceed the threshold, sites explain part of the residual, and $b_W - b_{\text{ref}}$ quantifies which part. All other combinations leave the accounts open. Neither contrast establishes which property of a site matters, and neither account excludes the other outside the tested range.

5 Results

Thirteen of the 30 cancer types meet the site-pool requirement in all three splits, above the minimum of ten, and all 45 fits completed. The multi-site classes are the larger TCGA-UT cancer types listed in Table 5, including breast, lung, colon, and prostate carcinomas and both gliomas. Before accuracy is restricted to multi-site classes, refitting the effective-support regression to all 30 classes reproduces the full-feature residual of 2.62 points per doubling reported by Queisler (2026b).

5.1 Allocation accuracy

Most of the accuracy difference between allocations arises in the first step. Quadrupling patients within the same five sites raises accuracy on multi-site classes from 56.36 to 69.88 percent, a gain of 13.52 points (95% interval: 13.05 to 14.03; Table 3). Replacing ten of these twenty patients with patients from five further sites raises it by a further 1.06 points, to 70.94 percent. The deep allocation is also the most sensitive to which patients are drawn: its accuracy varies with a standard deviation of 2.34 points across split–draw combinations, compared with 1.53 and 1.56 points for the two broad allocations.

Allocation	G	m	S	Multi-site classes		All classes
				Accuracy [95% interval]	SD	Accuracy [95% interval]
Deep	5	32	5	56.36 [55.18, 57.57]	2.34	65.01 [63.77, 66.22]
Broad, five sites	20	8	5	69.88 [68.62, 71.17]	1.53	68.50 [67.19, 69.74]
Broad, ten sites	20	8	10	70.94 [69.71, 72.21]	1.56	68.86 [67.54, 70.12]

Table 3: Adding patients within fixed sites raises accuracy on multi-site classes far more than adding sites at a fixed patient count. G , m , and S are patients per class, patches per patient, and sites per class for multi-site classes. Accuracy is patient-macro balanced accuracy in percent, averaged over three splits and five draws, with test-patient 95% intervals. SD is the population standard deviation of accuracy on multi-site classes across the fifteen split–draw combinations, in percentage points.

Accuracy over all 30 classes changes less, because the seventeen background classes receive identical training material in every allocation. Subtracting the contribution of multi-site classes from

the all-class average gives a background-class accuracy of 71.61 percent for the deep allocation and of 67.43 and 67.27 percent for the two broad allocations. The 4.18-point decrease in the first step is consistent with multi-site classes attracting more predictions once they are trained on more patients. The site step changes background-class accuracy by only 0.16 points.

5.2 Site gain and within-site residual breadth

Neither contrast falls in the region that the interpretation criteria assign to a single account (Table 4). The site gain is $\Delta_S = 1.06$ points with a test-patient interval of 0.74 to 1.37. The interval excludes zero but crosses the one-point threshold, so under test-patient uncertainty the gain from five additional sites is positive but not established as practically relevant.

Quantity	Symbol	Estimate	95% interval
First step, raw	$A_5 - A_D$	13.52	[13.05, 14.03]
Effective-support slope	$\hat{\beta}$	5.37	[4.63, 6.14]
First step explained by effective support	$\hat{\beta} \log(N_{\text{eff}}(20, 8)/N_{\text{eff}}(5, 32))$	6.90	[5.94, 7.89]
Within-site residual breadth	b_W	3.31	[2.71, 3.84]
Patient-count slope	$\hat{\gamma}$	1.75	[0.91, 2.58]
Reference residual breadth	b_{ref}	1.21	[0.63, 1.79]
Within-site minus reference	$b_W - b_{\text{ref}}$	2.10	[1.77, 2.40]
Site gain	Δ_S	1.06	[0.74, 1.37]

Table 4: The within-site residual lies above the one-point threshold, whereas the site gain interval crosses it. All quantities are restricted to the thirteen multi-site classes. Accuracy differences and residuals are in percentage points, residuals per doubling of patients; slopes are in percentage points per unit of natural logarithm. Intervals are paired test-patient 95% intervals.

For the within-site residual, effective support of the multi-site classes rises from 10.36 to 37.44 per class between the deep and the broad five-site allocation. The refitted slope $\hat{\beta} = 5.37$ attributes 6.90 of the 13.52-point first step to this increase. The remaining 6.62 points, spread over two doublings of patients, give $b_W = 3.31$ points per doubling (2.71 to 3.84). This interval lies entirely above the threshold, although the comparison with the breadth–depth grid below attributes most of this value to the deep allocation.

The reference residual for the same classes is smaller. With sites uncontrolled, the refitted regression gives $b_{\text{ref}} = 1.21$ points per doubling (0.63 to 1.79), less than half of the 2.62 points obtained over all 30 classes. The within-site residual exceeds it by 2.10 points (1.77 to 2.40). Equivalently, the regression predicts a first step of 9.33 points for randomly drawn allocations of the multi-site classes, whereas the site-restricted allocations show 13.52 points.

The excess lies in the deep allocation rather than in the within-site step. On the stored grid predictions restricted to the same multi-site classes, the step from five patients with 32 patches to twenty patients with eight is 9.30 points (8.73 to 9.92), within 0.03 points of the regression prediction. The five-site allocation matches its grid counterpart within 0.52 points (0.02 to 1.03), whereas the deep allocation falls 3.70 points (3.11 to 4.27) below randomly drawn five-patient allocations, in 13 of the 15 split–draw combinations. With the grid step in place of the allocation step, the residual is 1.20 points per doubling, equal to b_{ref} . Twenty patients from five sites are thus at least as accurate as twenty randomly drawn patients, and b_W mainly reflects a weak deep allocation rather than a larger benefit of patients from already represented sites.

The test-patient intervals hold the sampled sites and patients fixed. Across the fifteen split–draw combinations, the two steps differ in how consistently they help. The raw first step is positive in all fifteen, ranging from 7.13 to 17.12 points with a standard deviation of 2.45. The site gain ranges from -1.86 to 4.96 points with a standard deviation of 1.70 and is positive in

eleven of fifteen combinations. Part of this spread lies between splits, whose mean site gains are 0.39, 0.72, and 2.06 points. Whether five additional sites help at all therefore depends on which sites and patients are drawn, whereas the benefit of additional patients within fixed sites does not.

Under Table 2, the within-site residual alone matches both the within-site variation and the combined account. The site gain matches neither, because its interval lies neither entirely above one point nor entirely within $[-1, 1]$. The outcome is therefore inconclusive. Because the deep allocation inflates b_W , its position above the threshold does not by itself establish a within-site benefit of that size.

5.3 Site strata

The site gain is concentrated on test patients whose sites enter only the ten-site allocation (Table 5). Pooled with the patient weights of the primary endpoint, the gain for patients from added sites is 3.93 points and is positive in all thirteen classes, ranging from 0.26 to 10.09 points. Patients from core sites, whose sites lose two of their four training patients, lose 1.53 points; the change is negative in eleven classes. Patients from sites drawn for neither allocation gain 0.67 points, with positive changes in nine classes. Added-site and core-site patients each make up about a third of the test patients of multi-site classes, and patients from other sites slightly more. Weighted by these shares, added-site patients account for 1.26 of the pooled 1.04-point gain, core-site patients for -0.48 , and other patients for 0.25.

Multi-site class	Shares, % core / added / other	Site gain by stratum			
		Core	Added	Other	Class
Bladder urothelial carcinoma	30 / 25 / 45	-4.18	7.03	-1.00	0.07
Brain lower grade glioma	30 / 30 / 40	-2.57	2.04	-1.72	-0.85
Breast invasive carcinoma	25 / 26 / 49	-1.51	0.53	0.54	0.02
Colon adenocarcinoma	29 / 42 / 29	-0.46	8.64	1.62	3.97
Glioblastoma multiforme	48 / 46 / 6	-1.15	2.02	2.31	0.51
Head and neck squamous cell carcinoma	42 / 34 / 24	-1.38	0.26	3.74	0.40
Kidney renal papillary cell carcinoma	29 / 29 / 42	0.75	1.15	-0.42	0.37
Lung adenocarcinoma	19 / 27 / 54	-0.31	5.27	0.81	1.80
Lung squamous cell carcinoma	21 / 24 / 55	-3.86	10.09	1.37	2.34
Prostate adenocarcinoma	31 / 38 / 32	0.13	2.51	0.17	1.04
Skin cutaneous melanoma	36 / 33 / 31	-2.16	5.03	-0.01	0.91
Thyroid carcinoma	38 / 30 / 31	-1.03	0.55	0.10	-0.19
Uterine corpus endometrial carcinoma	27 / 34 / 38	-2.71	6.81	3.82	3.07
Pooled	31 / 32 / 37	-1.53	3.93	0.67	1.04

Table 5: Adding sites helps mainly test patients from the added sites. Shares give the percentage of a class’s test patients from core, added, and other sites. Site gains are ten-site minus five-site recall in percentage points, averaged over the fifteen split–draw combinations. The class column combines the strata with their shares. Pooled rows weight classes equally, as in the primary endpoint. Because shares and recalls are averaged separately over split–draw combinations, the pooled class gain of 1.04 differs slightly from $\Delta_S = 1.06$. Estimates are descriptive and carry no intervals.

The class-level gains follow the added-site gains. The largest occur in colon adenocarcinoma, uterine corpus endometrial carcinoma, and lung squamous cell carcinoma, at 2.34 to 3.97 points, each with an added-site gain of at least 6.81 points. Lower-grade glioma and thyroid carcinoma lose accuracy, because losses for core-site patients, and for lower-grade glioma also for patients from other sites, outweigh their added-site gains.

6 Discussion and conclusion

For the thirteen larger TCGA-UT cancer types, holding sites fixed does not remove the benefit of additional patients. Twenty patients from five sites are at least as accurate as twenty randomly drawn patients, and on the breadth–depth grid the residual for these classes is 1.20 points per doubling of patients, close to the one-point threshold. The within-site residual of 3.31 points per doubling overstates this benefit, because most of it reflects a deep site allocation that is 3.70 points less accurate than randomly drawn five-patient allocations. Doubling the number of sites at a fixed patient count adds 1.06 points, about as much as a doubling of patients adds beyond effective support. This site gain changes sign in four of fifteen training draws, and its interval crosses the threshold, so the formal outcome remains inconclusive. Neither account dominates: additional sites and additional patients each contribute about one point, and neither contribution is established as practically relevant.

The strata indicate what this site contribution consists of. The gain falls on test patients from sites that the ten-site allocation has seen, while patients from core sites lose part of the benefit of their four training patients per site. This is the pattern predicted by site matching. Patients from sites drawn for neither allocation, the stratum closest to deployment at an unseen institution, gain 0.67 points. Within the tested range, broader institutional coverage at a fixed patient count therefore offers limited robustness to new institutions. Site-aware sampling would be most useful when the deployment sites are known and can be represented in training.

The within-site residual exceeds the reference residual by about two points, whereas the interpretation criteria expected a value between zero and b_{ref} . Three differences between the two estimates could produce such a gap, and the comparison with the grid narrows them. First, b_{ref} summarizes the patient-count slope across nine allocations, whereas b_W describes a single step from five patients with 32 patches to twenty patients with eight patches; curvature in how accuracy depends on patients and depth would separate the two. The raw grid step matches the regression prediction within 0.03 points, which rules this out. Second, the correlations ignore any similarity of patients from the same site. If such similarity exists, effective support of the broad five-site allocation, with four patients per site, is overstated relative to the deep allocation, with one patient per site. The explained part of the first step would then be too large, and b_W would understate the within-site residual, so this bias cannot explain the gap either. Third, the site pool admits only patients from institutions with at least four eligible patients, so site-restricted allocations draw from a narrower patient population than the random allocations of the grid. The gap arises entirely at the deep end: the five-site allocation matches its grid counterpart, whereas the deep allocation, one patient from each of five pool sites, does not. Drawing five patients from the site pool therefore lowers accuracy, whereas drawing twenty does not. The experiment does not identify which property of these patients or sites causes this.

The multi-site classes also carry less residual breadth benefit than the full label space: 1.21 points per doubling, less than half of the 2.62 points across all 30 classes, and consistent with the grid-based residual of 1.20 points. Most of the unexplained benefit of additional patients thus lies in the seventeen smaller cancer types, which lack enough patients per site for this intervention. Whether site coverage explains the residual for these types, which are most exposed to patient shortage, remains untested.

Within these limits, restricting training patients to few institutions costs these cancer types little accuracy, and additional patients still contribute about one point per doubling that effective support does not capture. Because the site gain mainly reflects matching of represented sites rather than robustness to unseen ones, a measurable shortage signal for these cancer types should quantify how much of the within-class variation between patients the training cohort covers, rather than how many institutions it represents. Testing this requires a within-class measure of patient coverage that does not depend on site labels, together with a design that reaches the smaller cancer types in which most of the residual breadth benefit arises.

7 Limitations

A tissue source site bundles preparation, staining, patient population, and referral patterns. A positive site gain would show that institutional variety helps, but not which of these components carries the benefit; separating staining from population would require stain normalization or site-matched biological covariates. Conversely, a null site gain applies to the tested step from five to ten sites and does not exclude benefits from a much wider site range.

The site pool restricts the contrast to sites with at least four eligible patients and to classes with at least ten such sites. Multi-site classes are therefore larger cancer types collected by larger institutions. Rare cancer types, which are most exposed to patient shortage, enter only as fixed background classes, and conclusions may not transfer to them. Background classes are identical across allocations, but their recall can still change through confusions with multi-site classes, which is why balanced accuracy over all classes is reported alongside.

The within-site residual inherits the assumptions of the effective-support regression. Its correlations are estimated from patients of all sites. If patients of one site resemble each other more than patients of different sites, the regression overstates the effective support of the site-restricted allocations, and part of a site effect would appear in the first step as a smaller b_W rather than only in Δ_S . The regression slope is also estimated on randomly drawn allocations and transferred to site-restricted ones. The site pool changes the population from which allocations are drawn as well: the deep site allocation is less accurate than randomly drawn five-patient allocations, so comparisons between site-restricted and random allocations mix the effect of sites with that of the patients the pool admits. Finally, all results are conditional on frozen Virchow2 features, multinomial logistic regression, the three overlapping splits, and the tested allocation sizes.

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